

# ABHINANDAN DALAL

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## CONTACT INFORMATION

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 [abhinandan-dalal.github.io/website](#)

## ACADEMIC AFFILIATION

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- **Postdoctoral Research Fellow**, Department of Health Care Policy, Harvard University.  
Mentor: Alex Luedtke July 2026 - current
- **PhD in Statistics & Data Science**, The Wharton School, University of Pennsylvania  
Advisor: Dylan Small Fall 2021 - Spring 2026  
Dissertation: [Modern Advances in Accounting for Unmeasured Confounding via Sensitivity Analysis](#)
- **Master of Statistics**, Indian Statistical Institute Fall 2019 - Spring 2021
- **Bachelor of Statistics**, Indian Statistical Institute Fall 2016 - Spring 2019

## PUBLICATIONS AND PREPRINTS

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### Accepted/Published

1. **Abhinandan Dalal**, Iris Horng, Yang Feng, Dylan S. Small (2026). The Role of Measured Covariates in Assessing Sensitivity to Unmeasured Confounding. *arXiv:2602.14414*. Forthcoming at *Observational Studies*.
2. William Bekerman\*, **Abhinandan Dalal\***, Carlo del Ninno, Dylan S. Small (2025). Planning for Gold: Hypothesis screening with split samples for valid powerful testing in matched observational studies. *arXiv:2406.00866*. *Biometrika*.
3. Abhinav Chakraborty\*, Anirban Chatterjee\* and **Abhinandan Dalal\*** (2024). PrIsing: Privacy-Preserving Peer Effect Estimation via Ising Model. *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR 238:2692-2700.

### Manuscripts under review/revision

4. **Abhinandan Dalal** and Eric J. Tchetgen Tchetgen (2025). Partial Identification of Causal Effects for Endogenous Continuous Treatments. *arXiv:2508.13946*.
5. **Abhinandan Dalal**, Patrick Blöbaum, Shiva Kasiviswanathan and Aaditya Ramdas (2024). Anytime-Valid Inference for Double/Debiased Machine Learning of Causal Parameters. *arXiv:2408.09598*.

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\* denotes equal contribution

## Technical Reports

1. **Abhinandan Dalal**, Diganta Mukherjee and Soumendu Sundar Mukherjee (2021). An endogenous (game-theoretic) approach to explore epidemic dynamics on a network (Masters Dissertation). [Link to report](#)
2. Navonil Deb\*, **Abhinandan Dalal**\* and Gopal K Basak (2020). Finding Optimal Cancer Treatment using Markov Decision Process to Improve Overall Health and Quality of Life. *arXiv:2011.13960*
3. Arindam Roy Chowdhury\*, **Abhinandan Dalal**\* and Shubhajit Sen\* (2020). Accelerography: Feasibility of Gesture Typing using Accelerometer. *arXiv:2003.14310*
4. Diganta Mukherjee, **Abhinandan Dalal** and Subhrajyoty Roy (2019). Feasibility of Transparent Price Discovery in Tea through Auction in India. *Commodity Insights Yearbook 2019, Multi Commodity Exchange of India Ltd.* [Link to article](#)

## AWARDS AND ACCOLADES

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| <b>June 2026</b> | <b>Outstanding Presentation Award</b> , 2026 NISS New Researchers Conference                                                                                                                                                                                                |
| <b>Oct 2025</b>  | <b>Lawrence D. Brown PhD Student Paper Award</b> , University of Pennsylvania                                                                                                                                                                                               |
| <b>Sept 2025</b> | <b>J. Parker Bursk Award for Excellence in Research</b> , University of Pennsylvania                                                                                                                                                                                        |
| <b>Sept 2025</b> | <b>Donald S. Murray Prize for Excellence in Teaching</b> , University of Pennsylvania                                                                                                                                                                                       |
| <b>June 2025</b> | <b>Best Student Paper in Application of Statistics and Data Science</b> , International Indian Statistical Association                                                                                                                                                      |
| <b>May 2025</b>  | <b>George James Term Fund for American Causal Inference Conference</b> , University of Pennsylvania                                                                                                                                                                         |
| <b>Dec 2024</b>  | <b>Nominated for Best Student Paper in Application of Statistics and Data Science</b> , International Indian Statistical Association                                                                                                                                        |
| <b>May 2024</b>  | <b>George James Term Fund for International Conference on Artificial Intelligence and Statistics (AISTATS)</b> , University of Pennsylvania                                                                                                                                 |
| <b>June 2022</b> | <b>George James Term Fund for The 2022 IMS Annual Meeting</b> , University of Pennsylvania                                                                                                                                                                                  |
| <b>May 2021</b>  | <b>Nominated for Prasanta Chandra Mahalanobis Gold Medal</b> , Indian Statistical Institute                                                                                                                                                                                 |
| <b>Dec 2019</b>  | <b>Nominated for best paper at International Conference of Operations Research and Decision Sciences (ICORDS)</b> , Indian Institute of Management, Vishakhapatnam                                                                                                          |
| <b>Mar 2019</b>  | <b>All India Rank 1</b> in Mathematical Statistics (MS) in the Joint Admission Test for M.Sc. 2019 (JAM) conducted by the Indian Institute of Technology(s)                                                                                                                 |
| <b>Dec 2018</b>  | <b>top 25 percentile in the world “Individuals category” and best individual entry</b> from ISI Kolkata at <b>Simon Marais Mathematics Competition</b> undergraduate <b>Asia-Pacific Mathematics</b> competition inspired by William Lowell Putnam Mathematical Competition |
| <b>2016 - 21</b> | <b>Dean’s List for Toppers</b> , Indian Statistical Institute                                                                                                                                                                                                               |
| <b>June 2016</b> | Secured WBJEE (West Bengal Joint Entrance Examination) rank <b>53</b> in engineering entrance                                                                                                                                                                               |

## INTERNSHIP EXPERIENCE

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- May 2024**    **Applied Scientist Intern**, SCOT Regional Forecasting, **Amazon**  
*Feasibility and utility analysis of Earth Mover distance optimization for regional demand forecasting using proximal causal analysis. Analysis predicts an estimated annual savings of 78million USD.*                      Supervisors: Rahul Gopalsamy & Kenny Shirley
- May 2023**    **Applied Scientist Intern**, **Amazon Web Services**  
*Developed anytime valid inference for causal effect estimation under double/debiased machine learning framework*    Supervisors: Patrick Blöbaum & Shiva Kasiviswanathan
- May 2020**    **Summer@Duke Statistical Science internship** (virtual due to Covid 19)  
*On Designing powerful randomization for exact p-values of causal effects under network interference*                      Supervisor: Alexander Volfovsky
- May 2020**    **Research Intern at Bio-inspired Communication (BCS) Lab, Technische Universität (TU) Darmstadt**, Germany (virtual due to Covid 19)  
*On Moments of Causal Estimator of binary Markovian input given output in a Poisson channel*                      Supervisor: Heinz Köppl
- May 2019**    **Summer Intern at DST- Center for Policy Research at Indian Institute of Science**  
*Analysing usefulness of public R & D grants in India- BIRAC and SBIR: Exploring Causal Aspects using Regression Discontinuity Design*                      Supervisor: Venkat Nadella

## TEACHING AND MENTORING

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### Teaching Assistantships at Penn

- Fall 2025**        STAT 9210: **Observational Studies** (for Graduate Students)
- Spring 2025**    STAT 5330: **Stochastic processes** (for Undergraduate Students)
- Spring 2025**    STAT 1110: **Introductory Statistics** (for Undergraduate Students)
- Fall 2024**        STAT 4320: **Mathematical Statistics** (for Undergraduate Students)
- Fall 2023**        STAT 4300 & STAT 5100: **Probability** (for Undergraduate/Graduate Students)
- Spring 2023**    STAT 9700: **Mathematical Statistics** (for Graduate Students)
- Spring 2022**    STAT 521: **Applied Econometrics II** (for Graduate Students)

### Mentoring Experience

- Spring 2025**    **Wharton Directed Reading Program (WDRP)** mentor for *Sheyan Lalmo-hammed*
- Fall 2024**        **PhD Peer program** mentor for incoming PhD student *Soham Mallick* to navigate the program and transition to Penn
- Spring 2024**    **Undergraduate Research in Probability and Statistics (URPS)** mentor for *Roberto Ligeralde*, with Da Wu
- Spring 2022**    **Undergraduate Research in Probability and Statistics (URPS)** mentor for *Ryan Jeong* and *Alexandru Lopotenco*, with Prof. Robin Pemantle

## CONFERENCES & WORKSHOPS

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- Partial Identification of Causal Effects for Continuous Treatments under Sensitivity Models
  - NISS Virtual New Researchers Conference (2026).
  - American Causal Inference Conference, Salt Lake City, UT (2026).
  - Lawrence D. Brown PhD Student Workshop, University of Pennsylvania, Philadelphia, PA (2025) **Invited Talk**.
  - Penn Center for Causal Inference Seminar, University of Pennsylvania, Philadelphia, PA (2025).
- Observational Causal Inference with Unknown/Mismeasured Network Interference
  - NetSci Conference, CausNets Satellite, Boston, MA (2026).
  - PhD Student Seminar, Dept. of Statistics & Data Science, University of Pennsylvania, Philadelphia, PA (2025).
- Planning for Gold: Hypothesis screening with split samples for valid powerful testing in matched observational studies.
  - Annual Conference of International Indian Statistical Association, Lincoln, NE (2025).
  - PhD Student Seminar, Dept. of Statistics & Data Science, University of Pennsylvania, Philadelphia, PA (2024).
- Role of Earth-Movers Distance as a metric for Regional Forecasting
  - Intern Presentations, Amazon Supply Chain Optimization Technologies, New York City, NY (2024).
- Anytime Valid Inference for Double/Debiased Machine Learning of Causal Parameters
  - Conference on Frontiers in Machine Learning and Econometrics, co-hosted by the Federal Reserve Bank of Philadelphia and Center for Applied AI at the University of Chicago Booth School of Business, Chicago, IL (2025).
  - American Causal Inference Conference, Detroit, MI (2025).
  - Annual Conference of International Indian Statistical Association, Kochi, India (2024).
  - Penn Center for Causal Inference Seminar, Philadelphia, PA (2024).
  - Young Researcher Presentations, Amazon Supply Chain Optimization Technologies, New York City, NY (2024).
  - Intern Presentations, Amazon Web Services, Santa Clara, CA (2023).
- Modeling county level firearms related fatal assaults in the United States
  - Joint Statistical Meeting, Toronto, Canada (2023).
- An endogenous approach to explore epidemic dynamics on a network
  - IMS Annual Meeting, London, UK (2022).
- Modern Statistical Challenges under a Causal Lens
  - P.C.M. Gold Medal Presentation, Indian Statistical Institute, Kolkata, India (2021) **Invited Talk**.

- Feasibility of Automation in Price Setting for Tea Auction
  - International Conference of Operations Research and Decision Sciences (ICORDS), Indian Institute of Management Vishakhapatnam, India (2019).
  - XXIXth Annual General Conference on Contemporary Issues in Development Economics, Jadavpur University, Kolkata, India (2019).
  - Third International Conference on Issues in Economic Theory and Policy, Presidency University, Kolkata, India (2019).

## PROFESSIONAL SERVICE

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- **Reviewer** for Journal of the Royal Statistical Society: Series B, Biometrics, Journal of the American Statistical Association, Computational Statistics, Applied Stochastic Models in Business and Industry
- **Student Member** Wharton Causal Data Science Lab ([WCDSL](#)), Penn Center for Causal Inference ([CCI](#))
- **Co-organizer** for Statistics Dept. Student Seminar 2021-22 at Penn

## SKILLS

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### Computer Skills

- R, MS Office
- $\text{\LaTeX}$
- C, SPSS, SQL, Git (basic)

### Language Skills

- Bengali (native)
- English, Hindi (fluent)
- Spanish (rudimentary)